

and the United Arab Emirates expect digital assets will replace fiat currencies by almost 90 per cent in the next 5 to 10 years. Respondents from Singapore, Ireland, and Israel say the replacement will be 80 per cent. Apart from tax and regulatory concerns about digital asset usage, 66 per cent of respondents say that cybersecurity issues are preventing widespread implementation.

Based on their inherent characteristics, blockchains are classified as public, private and consortium (hybrid) blockchains (Table 1).

In the crypto industry, the public blockchain is the most well-known and Ethereum the most popular. Solidity is Ethereum's own coin and programming language. It is commonly employed in the fields of business, finance, and entertainment. Quorum is an open source technology that is used in private blockchain networks, where a single node is held by a single member of the network. Quorum is a revised version of Ethereum where the Geth client is modified. Corda is a permission based blockchain platform which uses DLT applications. Its transactions are strictly private and it provides multi-party transparency. Corda, Quorum and Hyperledger Fabric are referred to as enterprise blockchains with data privacy as their topmost priority. Ethereum enjoys an independent role for specific applications. Hyperledger Fabric is well known for its modular and extendable architecture in various companies and Corda has wide acceptance in the financial services industry.

Distributed ledger technology (DLT) provides a backup to facilitate replication of data. The ledger, an integral distributed branch, is shared and synchronised across multiple platforms with simultaneous access for multiple users. However, once a piece of data gets logged to a ledger it can neither be edited (immutability) nor

Features	Public	Private	Consortium
Participation	Without permission	By permission	
Centralisation	None	Partial	
Member	Anyone	By invitation	Membership with partial access
Ledger	Open	Selective openness	
Security	Weak	Controlled by administrator	
Consensus	Proof of work	Variable consensus	
Legal ownership	None	Owner	Selected nodes
Tampering	None	Members can collaborate to alter ledger	None
Computation	High	Medium	Medium
Network dimension	Large	Small	Medium
Examples	Bitcoin, Litecoin, Ethereum	Corda, Hyperledger Fabric, Quorum	Ripple, IBM Food Trust

Table 1: Classification of blockchain

Properties	Description
Anonymous	Identity of participants is kept pseudonymous
Distributed	Every participant of a network has a copy of a ledger to ensure transparency
Immutable	Validated records are irreversible and cannot be changed
Programmable	A blockchain is programmable (such as smart contracts)
Secure	Encryption of all records is made individually
Timestamped	A transaction timestamp is recorded on a block
Unanimous	Every participant of a network agrees to the validity of each of the records

Table 2: Properties of DLT

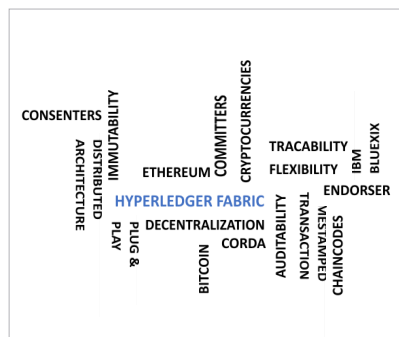


Figure 2: Salient features of blockchain, DLT and Hyperledger Fabric

be removed from the database. This is how the system design is executed. Properties of DLT that enable the blockchain to operate in a modular manner are listed in Table 2.

Programming languages for smart contracts in distributed ledger technology are listed in Table 3. Each has at least one unique characteristic.

Hyperledger Fabric

A modular blockchain framework known as Hyperledger Fabric acts